Q.21 What are the functions of flow switches?

Q.22 Why temperature switch is used in the industries? Explain with example.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Write and explain all the important components of pnematic control system. Differentiate between hydraulic and pneumatic control system

Q.24 Draw and discuss the working of cascade control system with example. How does a ratio control system special case of cascade control?

Q.25 Find the transfer function of PI controller. Explain the integral action using input-output curve. Write down the advantages and disadvantages of it.

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Roll No. ....

4th Sem / Instrumentation & Control

Subject : Process Control

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The system which has only one manipulated variable and more than one measurement is

a) Feed-forward control system

b) Cascade control system

c) Ratio control system

d) Split-range control system

Q.2 The control used to reduce both rise time and steady-state error is

a) P controller b) D controller

c) P-D controller d) P-I controller

Q.3 Which of the following component is used for pneumatic pressure supply

a) Cylinder b) Air compressor

c) Pneumatic valve d) Actuator

Q.4 A pneumatic system generally operates on

a) (5-50) bar pressure

b) (3-15) bar pressure

c) (5-10) bar pressure

d) Above 100 bar pressure

Q.5 The ______ of a diaphragm valve operates the diaphragm.

a) Bonnet b) Stem

c) Compressor d) Pilot channel

Q.6 The permanent magnet is used in the flow switch because

a) It is fastened to the paddle's top and activated to activate a reed switch, producing a potential-free output.

b) It is used as interlocking circuit

c) It is used as sensing element

d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A_____ control system is a closed loop system in which multiple controllers are used to read multiple input values.

Q.8 Split range control is most commonly used in flare application of refinery and chemical plants. (True/False)

Q.9 Define Intergral controller.

Q.10 The input of flapper--nozzle system is (4-20) mA signal and the equivalent output is (3- to 15) psi pressure.(True/False)

Q.11 Define the globe valve.

Q.12 Define the pressure switch.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What are the applications of solenoid valve?

Q.14 Explain the construction details of globe valves.

Q.15 Explain the working of feed-forward control.

Q.16 Write any four difference between feed-forward control and conventional feedback control system?

Q.17 Write any two advantages and two advantages of PD controller.

Q.18 Draw and explain the I/P coverter using flapper-nozzle system.

Q.19 What is solenoid valve? How does it work?

Q.20 Explain the construction details of ball valves.